

APIs and data sources

Every external service and internal endpoint used by the BUMBLEBLE thunderstorm nowcasting system, what each one provides, where it is used, and what it costs.

Prepared for the Smart India Hackathon review, October 2026.

Summary

BUMBLEBLE runs in **real-time mode** on free cloud APIs only, so it can be deployed to any host (for example Hugging Face Spaces or Render) without accounts, file uploads or paid services. Four data APIs feed the forecast and the map: **RainViewer** for weather radar, **NASA GIBS** for Himawari-9 infrared satellite images, and **Open-Meteo** for surface weather, wind and place search. Map backgrounds come from **Esri** and **OpenTopoMap**. The only service that needs a key is the optional **Google Gemini** AI assistant, which has a free tier; without a key the assistant runs in a built-in demo mode.

API	Provides	Called from	Key needed	Cost
RainViewer Weather Maps	Weather radar (rain and storm intensity), every 10 min	Server + browser	No	Free
NASA GIBS WMS	Himawari-9 infrared cloud-top temperature, every 10 min	Server	No	Free
Open-Meteo Forecast	Current weather, next hours, wind field	Server + browser	No	Free (non-commercial)
Open-Meteo Geocoding	Place search worldwide	Browser	No	Free (non-commercial)
Esri ArcGIS basemaps	Map backgrounds (default, streets, satellite, terrain)	Browser	No	Free with attribution
OpenTopoMap	Topographic map background	Browser	No	Free (CC-BY-SA)
Google Gemini API	AI assistant answers and document search	Server	Yes (optional)	Free tier
Anthropic Claude API	Alternative AI assistant	Server	Yes (optional)	Paid

Libraries and fonts are loaded from public CDNs (Leaflet from cdnjs, Inter and JetBrains Mono from Google Fonts). They are not data APIs, so they are listed in section 4 only.

1. Live weather data

These APIs supply the real observations behind the forecast. The server pulls radar and satellite data in the background, merges them on a 2 km grid over East and North-East India, and forecasts 0–6 hours ahead.

RainViewer Weather Maps API — weather radar

Purpose	Main input to the forecast. Shows where it is raining and how intense each storm is (reflectivity). Storm cells, their tracks and arrival countdowns are all built from it.
Endpoint	https://api.rainviewer.com/public/weather-maps.json lists the latest frames; radar tiles come from <code>{host}/{path}/256/{z}/{x}/{y}/2/...png</code>
Used in	Server: <code>backend/nowcast/sources/rainviewer.py</code> downloads zoom-7 tiles (~1.1 km per pixel), converts the colours back to dBZ with RainViewer's published colour table and resamples them onto the grid. Browser: the Rain radar loop animation on the map plays the last ~2 hours of frames worldwide.
Update rate	New frame every 10 minutes; the server checks every 60 seconds.
Access	Free, no key. Attribution "Radar © RainViewer" is shown on the map.
Limits	Reflectivity only (no Doppler wind), so radar-measured outflow is not available. Coverage depends on which national radars RainViewer receives.

NASA GIBS (Global Imagery Browse Services) — Himawari-9 infrared satellite

Purpose	Cloud-top temperature. Very cold tops (below about -50°C) mark tall thunderstorm clouds; tops that cool quickly warn of new storms before radar sees rain. Also fills gaps where radar has no coverage.
Endpoint	https://gibs.earthdata.nasa.gov/wms/epsg4326/best/wms.cgi — WMS GetMap, layer <code>Himawari_AHI_Band13_Clean_Infrared</code> . Colour key: <code>.../colormaps/v1.3/Clean_Longwave_Infrared_Window_Band.xml</code>
Used in	Server: <code>backend/nowcast/sources/himawari.py</code> . The image is requested directly on the analysis grid; every colour is turned back into a temperature using NASA's published colour key (1 $^{\circ}\text{C}$ bins). Images are corrected for parallax from Himawari-9's position at 140.7°E .
Update rate	One scan every 10 minutes, published about 40–90 minutes after the scan. The server checks every 5 minutes.
Access	Free, no key or account (NASA open data).
Limits	About an hour behind real time. The infrared window channel only; no water-vapour channel. Replaces INSAT-3DR, which requires a MOSDAC account.

Open-Meteo Forecast API — surface weather and wind

Purpose	The weather people feel right now: temperature, feels-like, humidity, wind, gusts, rain, conditions (clear, cloudy, thunderstorm...) and storm energy (CAPE), plus the next hours.
Endpoint	<code>https://api.open-meteo.com/v1/forecast</code> with <code>current=...</code> and <code>hourly=...</code> ; several locations are fetched in one request.
Used in	Server: <code>backend/nowcast/weather.py</code> powers BUMBLEBLE's <code>/api/weather</code> (any point on Earth) and <code>/api/weather/places</code> (all watched places), shown in the map pop-up and on the Arrivals page. Responses are cached for 10 minutes so many visitors share one call. Browser: the Wind flow animation fetches a 16 × 12 grid of current winds for the area on screen.
Update rate	Current conditions update about every 15 minutes.
Access	Free for non-commercial use, no key. Data licensed CC BY 4.0 (attribution shown).
Limits	Fair-use limit of about 10,000 calls per day per user on the free tier. Values come from weather models, not direct measurements.

Open-Meteo Geocoding API — place search

Purpose	Turns a typed place name into coordinates, so the search box finds any town in the world.
Endpoint	<code>https://geocoding-api.open-meteo.com/v1/search?name=...&count=10</code>
Used in	Browser: the map search box (<code>frontend/map.js</code>). The 45 watched places are matched instantly without a network call; other results come from this API, with Indian places ranked first.
Access	Free, no key.

2. Map backgrounds

Map tiles are loaded by the browser through the Leaflet map library and chosen from the **Map type** menu.

Map type	Service	Tile URL	Terms
Default (light / dark)	Esri World Light / Dark Gray Canvas	<code>server.arcgisonline.com/.../Canvas/World_Dark_Gray_Base/...</code>	Free with attribution
Streets	Esri World Street Map	<code>.../World_Street_Map/MapServer/tile/{z}/{y}/{x}</code>	Free with attribution
Satellite, Satellite + labels	Esri World Imagery + reference layers	<code>.../World_Imagery/...</code>	Free with attribution
Terrain	Esri World Terrain Base	<code>.../World_Terrain_Base/...</code>	Free with attribution
Topographic	OpenTopoMap	<code>{s}.tile.opentopomap.org/{z}/{x}/{y}.png</code>	CC-BY-SA, © OpenStreetMap

3. AI assistant

Google Gemini API (Google AI Studio)

Purpose	Powers the Ask AI page: answers questions in plain language using the live forecast, and searches the knowledge base (safety advice, IMD definitions, dashboard help) to cite its sources.
Models	gemini-2.5-flash by default, with automatic fallback to other Flash models if one is busy or retired; gemini-embedding-001 for knowledge-base search.
Used in	Server: backend/nowcast/assistant.py and backend/nowcast/rag.py.
Access	Needs a free API key from Google AI Studio, set on the server as GEMINI_API_KEY (never in the code or the repository). Free tier with rate limits.
Without a key	The assistant runs in demo mode : rule-based answers from the live data, so the site still works.

Anthropic Claude API (optional alternative)

Purpose	Same role as Gemini, used only if ANTHROPIC_API_KEY is set and no Gemini key is present.
Access	Paid, per use. Not needed for the free deployment.

4. Libraries and fonts (CDN)

Resource	Source	Use
Leaflet 1.9.4	cdnjs.cloudflare.com/ajax/libs/leaflet/1.9.4/	Interactive map
Inter, JetBrains Mono	fonts.googleapis.com, fonts.gstatic.com	Page fonts

5. Operational feeds (supported, not active)

The code can also read official Indian feeds. They need institutional access, so the free cloud deployment does not use them. Switch to them with `NOWCAST_MODE=live`.

Feed	Provider	Format	Access
Doppler weather radar	IMD (India Meteorological Department)	ODIM_H5 volumes (reflectivity + radial velocity)	Institutional request
INSAT-3D/3DR/3DS imager	ISRO MOSDAC	HDF5 (TIR1, water vapour)	Free registration
Lightning	IITM / IMD lightning network	Push to POST /api/ingest/lightning	By request

Lightning note: no free, official real-time lightning API exists for India. In real-time mode, lightning risk is estimated from radar intensity and cloud-top temperature; any network can push real strikes to the endpoint above.

6. BUMBLEBLE's own API

The FastAPI server exposes these endpoints to the web pages. All are read-only except where noted.

Method	Path	Returns / does
GET	/api/meta	Grid, places, forecast lead times, colour legends, radar sites
GET	/api/state	Latest analysis: storms, places at risk, alerts, feed health, accuracy
GET	/api/raster/{product}/{lead}.png	Map image for a layer (danger, lightning, hail, wind, rain...) at a lead time
GET	/api/point?lat=&lon=	6-hour hazard timeline for one spot in the forecast area
GET	/api/weather?lat=&lon=	Live weather now and next hours, anywhere on Earth (Open-Meteo)
GET	/api/weather/places	Live weather for every watched place (Open-Meteo)
WS	/ws	Live stream: new analyses, clock ticks, lightning strikes
GET	/api/assistant/info	Which AI provider and model are active
POST	/api/assistant	Ask the AI assistant a question
GET / POST / DELETE	/api/knowledge...	List, add, remove or search knowledge-base documents
GET	/api/export/bulletin.html	Printable bulletin
GET	/api/export/places.csv	Arrival times for all places (CSV)
GET	/api/export/nowcast.geojson	Storms, tracks and places for GIS tools
POST	/api/ingest/lightning	Push lightning strikes from an external network
POST	/api/sim/speed, /spawn, /source	Demo mode only: speed, add a storm, switch a feed off/on

7. Deployment checklist

- Deploy the included **Dockerfile** to a container host that supports WebSockets, such as Hugging Face Spaces (free) or Render.
- Set `NOWCAST_MODE=realtime` (the default). No data keys are needed.
- Optional: set `GEMINI_API_KEY` as a secret on the host to enable the AI assistant. Keep `.env` out of uploads; it is already excluded by `.gitignore` and `.dockerignore`.
- Static hosts such as Netlify or Vercel can show the pages but cannot run the forecast server.
- Keep the attributions visible on the map: RainViewer, Esri, OpenTopoMap/OpenStreetMap and Open-Meteo.
- All free APIs are for non-commercial use; check each provider's terms before any commercial launch.

Forecasts are guidance only. Always follow official IMD warnings.